



Incident report analysis

Summary	A multimedia company experienced a Denial of Service (DoS) attack that compromised its internal network, causing a two-hour service outage. The incident was caused by a malicious actor sending a flood of ICMP (ping) packets, which overwhelmed the network resources. This was made possible by an unconfigured firewall that lacked rules to filter or limit incoming ICMP traffic. The incident management team responded by blocking the incoming ICMP packets, taking non-critical services offline, and safely restoring critical network services. To prevent future occurrences, the security team implemented rate limiting, source IP verification, network monitoring software, and an IDS/IPS system.
Identify	Type of attack: Denial of Service (DoS) attack via an ICMP ping flood. Targeted assets: The company's internal network and all associated network services. Vulnerability: An unconfigured firewall that did not restrict, filter, or monitor the volume of incoming ICMP packets.
Protect	To further secure the organization's assets and prevent similar attacks, the following controls must be implemented: • Configure firewall rules to limit the rate of incoming ICMP packets (rate limiting) to prevent network exhaustion. • Enable Source IP address verification on the firewall to block spoofed IP addresses. • Implement an Intrusion Detection and Prevention System (IDS/IPS) to automatically filter and block malicious traffic based on known

	suspicious characteristics.
Detect	The organization must improve its visibility into network traffic by deploying centralized network monitoring software. This software should be configured to establish a baseline of normal network activity and generate automated alerts for the security team whenever abnormal traffic patterns are detected, such as sudden spikes in ICMP requests from external or untrusted IP addresses.
Respond	In the event of a future incident, the response plan should include: • Containment: Immediately block the malicious or spoofed source IP addresses at the perimeter firewall. • Neutralization: Reroute traffic or drop excessive ICMP packets using the IPS. • Isolation: Temporarily take non-critical services offline to preserve bandwidth and processing power for critical business operations until the attack is fully neutralized.
Recover	The recovery process involves a phased approach: • Restore and verify the functionality of critical network services first. • Gradually bring non-critical systems back online while closely monitoring the network for any lingering performance issues or secondary attacks. • Conduct a post-incident review to analyze logs, update firewall configurations, and ensure the network baseline is adjusted to handle similar threats without downtime in the future.

Reflections/Notes:

